

## **Milestone 02: Requirements & Backlog Expansion**

**CP317 - A - Group 6**

**Laurier Club Room Booking System**

### **Team Members:**

Product Owner: **Sawaab Anas**

Scrum Master: ***Muqadas Nazif***

Development Team: ***Manahil Bashir, Asmah Yasin Mohamed, Efetobore Salubi, Joleene Ismael***

## Requirements Refinement

The Laurier Clubs Room is meant to serve three primary people: club members and executives, space administrators and system administrators. Club users log in with their Laurier identity and land on a view that allows them to discover available rooms filtered by date, time and capacity and campus building. When the user adjusts filters, the system checks for any availability that might have conflict and then highlights the best fit rooms to the user. The user would also be given the ability to request a booking for a single event or a recurring form of events such as weekly meetings and be able to optionally add notes.

The system handles that when requests are submitted, it evaluates the constraints such as maximum advance days and double-booking rules, once it goes through it and gets approved, it would be given to the administrators for final confirmation if they want to approve, decline or request changes. In edge cases, administrators are able to handle conflicts where the system struggles and record it in an audit trail for clear visibility. Clubs can also review a booking history to track past and upcoming events, download confirmations and modify or cancel reservations subject to the policy listed. The system minimizes the collection of personal data and only requires info such as the name, student number, role, email and booking history and uses this to enforce role based access so that only authorized users can view or administer bookings.

### ***Functional Requirements (System Features)***

| <b><i>ID</i></b> | <b><i>Feature Name</i></b> | <b><i>Description</i></b>                                                                                  |
|------------------|----------------------------|------------------------------------------------------------------------------------------------------------|
| <b><i>F1</i></b> | User authentication        | Users log in with their Laurier Identity; roles define access(member, club executive, admin, system admin) |
| <b><i>F2</i></b> | Room search and filter     | Users search rooms by time, date, capacity and building; the system uses these to prevents conflicts       |
| <b><i>F3</i></b> | Booking request            | Users submit single or recurring room bookings with optional notes                                         |
| <b><i>F4</i></b> | Booking approval           | Admins approve, decline, or request changes for submitted bookings                                         |
| <b><i>F5</i></b> | Audit trail                | System logs all admin actions and conflict resolutions for transparency                                    |

|            |                          |                                                                                                                                             |
|------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>F6</b>  | Modify or cancel booking | Users can edit or cancel reservations within policies                                                                                       |
| <b>F7</b>  | Booking history          | Users view past and upcoming bookings                                                                                                       |
| <b>F8</b>  | Notifications            | Automatic alerts are sent for approvals, changes or cancellations, optional notifications are given as well for room spaces being freed up. |
| <b>F9</b>  | Policy management        | Admins set and enforce booking rules such as limits or advance days                                                                         |
| <b>F10</b> | Data privacy and access  | System stores minimal user data and restrict access by roles                                                                                |

### ***Non-Functional Requirements***

| <b><i>ID</i></b> | <b><i>Category</i></b> | <b><i>Description</i></b>                                               |
|------------------|------------------------|-------------------------------------------------------------------------|
| <b>NF1</b>       | Performance            | Search and booking actions should be completed fast                     |
| <b>NF2</b>       | Security               | Restricts access to unauthorized users                                  |
| <b>NF3</b>       | Usability              | Interface is clean, will be mobile friendly and accessible to all users |
| <b>NF4</b>       | Reliability            | System securely logs all transactions                                   |
| <b>NF5</b>       | Scalability            | Designed to support future campus or department expansion               |
| <b>NF6</b>       | Transparency           | All admin actions are tracked in an audit trail for accountability      |

## Ethical Considerations

### Ethical Considerations and How They Are Addressed:

1. Data Privacy of Students:

We solve this in the design by storing only the name, student number, role, email and booking history. This is done in order to ensure we do not take any other private information that they have.

2. Fairness and Equal Access:

To keep the system fair, we make sure no single club can take over all the rooms. We include limits on how many bookings a club can make in a week or how far in advance they can book. The website also shows clear booking rules so everyone knows what is allowed.

3. Security and Integrity:

We use Laurier's login system to keep all accounts protected. We make sure each user can only view or change what they are allowed to and non-users cannot access any information. This helps keep all data safe.

4. Accessibility and Inclusivity:

We want the system to be easy for everyone to use, including students with disabilities. The design will have clear text and simple navigation. This makes the website more inclusive and comfortable for all students.

5. Transparency and User control:

The system lets users always know what is happening with their booking. They can view their booking history, cancel or edit booking. If a booking is declined, be able to see the reason like quota exceeded. This helps users feel informed and in control.

6. Preventing Notification Overload:

We make sure users are not overwhelmed with unnecessary messages from the system. It only sends important updates, such as booking confirmation, declines or cancellations. This keeps notification simple and clear.

7. System Reliability and Trust:

We focus on building a reliable system that works smoothly especially during busy times. Reducing errors and downtime helps maintain user trust and prevent frustration.

8. Accountability and Compliance:

The system's design follows Canadian privacy and accessibility laws and AODA, and Laurier's own privacy and security policies. To ensure that the system is not only ethical but also meets proper legal standards.

# Initial Design Sketches

Login / Sign Up

## Laurier Club Booking

Email

Password

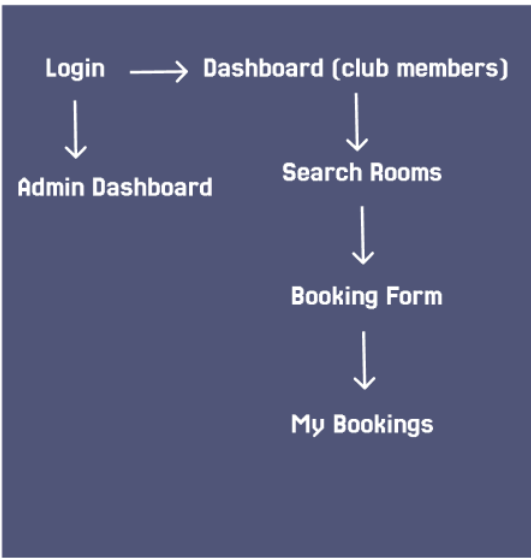
Role (Club / Admin)

Login

Sign Up

Forgot Password?

Navigation Map



Search Rooms

HomeMy BookingsBook Now

### Search Rooms

Building:

Room Type:

Capacity:

Date:

Time: —

Features:

Projector/Screen

Whiteboard/Blackboard

Computers

Microphone/Speakers

Book

Booking Form

HomeMy BookingsBook Now

Booking Room 1C10

Date: [ Oct 15, 2025 ]

Time: [ 3:00 PM - 5:00 PM ]

Purpose: [ Club Meeting ]

Recurring: [ Yes/No ]

Confirm Booking

Cancel

Admin Dashboard

[Home](#) [My Bookings](#) [Book Now](#)

MANAGE ROOMS

VIEW BOOKINGS

REPORTS

USERS

Pending Requests

Golden Hawks  
Room 1C10  
Oct 15: 3-5PM

Approve

Decline

Dashboard

[Home](#) [My Bookings](#) [Book Now](#)

Welcome, [User Name]!

Upcoming Bookings:

Room 1C10  
Oct 15: 3-5PM

Status: **Approved**

Room LH1001  
Oct 22: 4-7PM

(Resubmit)  
Status: **Declined**

Previous Bookings:

Room N1055  
Sep 25: 1-2PM

Status: **Completed**